

H-w Write a c++ code which includes a class student. Find out the space occupied by the object (size of)

```

1) #include <iostream>
using namespace std;
void main() {
    class student {
        int age r;
        String n;
        float m;
    public:
        void details (int r1, string n1, float m1) {
            r = r1;
            n = n1;
            m = m1;
        }
    }
    void main1() {
        student o;
        int r1;
        String n1;
        float m1;
        o.details (r1, n1, m1);
        print
        cout << sizeof(o);
    }
}

```

output:

	5
hai	
10	
48	

Here 48 is due to int → 4 bytes allocated as it is 64-bit compiler as it wants 8 bytes, so to get multiple 8 we add 4 bytes (ie padding)

int $\rightarrow$ 4 bytes

padding $\rightarrow$ 4 bytes

string $\rightarrow$ 32 bytes (64-bit compiler)

float $\rightarrow$ 4 bytes

44 bytes

it is not multiple of 8 so,

48 bytes

as 4 bytes add to 44 bytes

H.w Java generates Hash code for every object
find out what is hash code in java
and the purpose of generating hash
code.

Hash code in Java? It is a unique

integer value generated by Java for every object. It is returned by hashCode() method, which is define in the object class.

Purpose of Hash code:

- It helps Java identify and store objects efficiently.
- It is mainly used in collections like Hash map, Hash set, & Hashtable for fast Searching, insertion, & deletion.
- If two objects are equal according to the equals() method, they must have the same hash code.

Ex:

```
class Demo {
```

```
    public static void main (String[] args) {
```

```
        String s = "Hello";
```

```
        System.out.println("Hash code: " + s.hashCode());
```

```
    } }
```

Output: Hash Code: 69609650

NOTE: The hash code is an integer value. Different objects can sometimes have the same hashcode (called a hash collision) but Java handles such cases internally.

H.W
→ what are the various problems which we observed in Society (or) community. pick any 5 problems. Find out 'if any they

can be solve using data structure & Algorithms.

ans:

1. Traffic : Here we use dsa graphs which help to find the shortest & fastest route to reduce travel time.

2. Hospital Appointment Management: Here we use Queue which helps to manage patient appointments & prioritize emergency cases efficiently.

3. Waste collection Optimization : Here we use Graphs which helps to the plan the best route for garbage collection trucks to save time & fuel.

4. Library Book Management: Here we use the Searching Algorithms for quickly search, issue, & return books.

5. Emergency Ambulance Routing : To Find the nearest ambulance and the fastest path to the patient & Hospital using Graphs.